

A time-series analysis of any short-term effects of meteorological and air pollution factors on preterm births in London, UK

Sue J. Lee^{a,*}, Shakoor Hajat^a, Philip J. Steer^b, Veronique Filippi^a

^a*London School of Hygiene & Tropical Medicine, Keppel Street, London WC1E 7HT, UK*

^b*Academic Department of Obstetrics and Gynaecology, Faculty of Medicine, Imperial College London, 3rd Floor, Chelsea and Westminster Hospital, 369 Fulham Road, London SW10 9NH, UK*

Received 30 November 2006; received in revised form 26 September 2007; accepted 3 October 2007

Available online 19 November 2007

Abstract

Although much is known about the incidence and burden of preterm birth, its biological mechanisms are not well understood. While several studies have suggested that high levels of air pollution or exposure to particular climatic factors may be associated with an increased risk of preterm birth, other studies do not support such an association. To determine whether exposure to various environmental factors place a large London-based population at higher risk for preterm birth, we analyzed 482,568 births that occurred between 1988 and 2000 from the St. Mary's Maternity Information System database.

Using an ecological study design, any short-term associations between preterm birth and various environmental factors were investigated using time-series regression techniques. Environmental exposures included air pollution (ambient ozone and PM₁₀) and climatic factors (temperature, rainfall, sunshine, relative humidity, barometric pressure, and largest drop in barometric pressure). In addition to exposure on the day of birth, cumulative exposure up to 1 week before birth was investigated.

The risk of preterm birth did not increase with exposure to the levels of ambient air pollution or meteorological factors experienced by this population. Cumulative exposure from 0 to 6 days before birth also did not show any significant effect on the risk of preterm birth. This large study, covering 13 years, suggests that there is no association between preterm births and recent exposure to ambient air pollution or recent changes in the weather.

© 2007 Elsevier Inc. All rights reserved.

Keywords: Preterm birth; Air pollution; Environmental epidemiology; Time-series analysis; Weather; PM₁₀; Ozone

1. Introduction

Approximately 5–10% of all births in developed countries are preterm (Byrne and Morrison, 2004; Haram et al., 2003), defined as being born before 37 completed weeks of gestation (WHO, 1972). Although the routine collection of data on gestational age at birth does not occur at a national level in the United Kingdom (UK), generalization from local reports suggest that the preterm birth rate is about 7% (Chamberlain and Morgan, 2002). As

about 650,000 babies are born, on average, each year in England and Wales (Office for National Statistics, 2004), and approximately 1 in every 14 births are preterm, there will be over 45,000 preterm births each year.

Preterm birth is a syndrome (Romero et al., 2006) that is characterized by multiple etiology. It is therefore unusual to be able to identify an exact cause for each preterm birth. As such, preterm birth is not an adverse outcome in itself, but an important determinant of health status, specifically, neonatal morbidity and mortality (Villar et al., 2004). Despite dramatically improved survival rates in the past decade (Hack and Fanaroff, 1999), preterm birth remains one of the largest contributors to neonatal deaths (Copper et al., 1993; Goldenberg and Rouse, 1998). The economic costs of caring for babies who survive a preterm birth are very high (Petrou, 2003, 2005) and sequelae experienced

*Corresponding author. Mahidol-Oxford Tropical Medicine Research Unit, Faculty of Tropical Medicine, Mahidol University, 420/6 Ratchawithi Road, Ratchadewee, Bangkok 10400, Thailand.

Fax: +662 354 9169.

E-mail address: sue@tropmedres.ac (S.J. Lee).

later in life include lower educational achievement, decreased motor and cognitive functioning, and increased behavioral disorders, when compared with children who were born at term (Anderson and Doyle, 2003; Bhutta et al., 2002; Colvin et al., 2004; Foulder-Hughes and Cooke, 2003; Hack et al., 2002; Hediger et al., 2002; Marlow et al., 2005).

Although much is known about the incidence and burden of preterm birth, its biological mechanisms are not well understood. An important step in reducing preterm birth and its consequences is likely to lie in furthering our understanding of the complex and often inter-related pathways that lead to preterm birth. The role of environmental factors remains uncertain.

Current evidence that air pollution or meteorological factors may affect preterm birth is inconclusive. In addition, there are relatively few studies addressing associations between meteorological factors and preterm birth in the published literature (Lee, 2005). Aspects of weather that have been implicated previously in the initiation of early labor onset include changes in barometric pressure and exposure to high temperatures, although the findings are not consistent (Cesario, 2002; King et al., 1997; Lajinian et al., 1997; Noller et al., 1996; Porter et al., 1999). Similarly, studies on exposure to ambient levels of air pollution during pregnancy indicate conflicting results with regards to preterm birth. Studies reporting a positive association and studies reporting no association have been published (Glinianaia et al., 2004; Maisonet et al., 2004; Sram et al., 2005). In this study, we used data on all births that occurred at 15 of the 17 maternity units in the North West Thames region of London, UK, to investigate whether exposure to various environmental factors, including air pollution and climatic factors, affected the risk for preterm birth.

2. Methods

2.1. Outcome: preterm birth

Data on the gestational age of all live, singleton births ($n = 492,568$) from participating maternity units were collected as part of routine clinical practice and entered into the St. Mary's Maternity Information System (SMMIS) database from 1988 to 2000. The data were entered by midwives or trained clerks and programmed checks, such as online validation and standard definitions for clinical measurements, were in place to reduce the likelihood of implausible values and to ensure high quality data (Cleary et al., 1994). Plausibility checks to exclude impossible/implausible combinations of birthweight and gestational age were also conducted after the data were collected by inspecting scatterplots and checking the small number of outliers individually. Values that were more than three standard deviations of the mean birthweight for gestational age were considered unlikely, if the pregnancy was otherwise normal.

A gestational age of less than 24 weeks represents the limit for live birth registration in the UK (Tucker and McGuire, 2004); therefore, babies born before 24 weeks of gestation ($n = 456$) were not included in the analysis. Babies weighing 200 g or less were either associated with a gestational age of less than 24 weeks or considered an error ($n = 7$) and therefore also excluded. Finally, babies with congenital anomalies ($n = 9340$) were excluded, leaving 482,765 births for analysis.

A birth was defined as preterm if it occurred before 37 completed weeks of gestation (i.e., <259 days). In the SMMIS dataset, gestational age was primarily measured from the first day of the last menstrual period (LMP) (Ciconia Health Care Systems Limited, 1998), but if the woman had irregular menses or was uncertain of the date of her LMP, a 'best antenatal assessment' was made, taking into account data from first and second trimester ultrasound scans (which were universally offered and almost always accepted, during the study period). Individual level data (birth records) were collapsed so that the final dataset was a daily count of preterm births during the 13-year study period (1988–2000).

2.2. Exposure: climate

Daily data on six meteorological factors were obtained from the UK Met Office through the British Atmospheric Data Centre (<http://badc.nerc.ac.uk/home>; last accessed on 30 November 2006):

- maximum temperature (°C),
- minimum temperature (°C),
- rainfall amount (mm),
- amount of sunshine (hours),
- relative humidity (%), and
- barometric pressure (millibars (mbar)).

The daily mean values for relative humidity and barometric pressure were calculated by averaging 24 hourly values for each day. For barometric pressure, the largest daily drop was also derived from the 24 hourly measurements; days on which no drop in pressure occurred for the entire 24 h period were recorded as 0.

The collection site was situated at Heathrow airport. Range and consistency checks (e.g., with previous observations) were used to ensure the quality of the data each day.

2.3. Exposure: air pollution

To build the single pollutant models, daily measures were obtained for mean ambient levels of ozone and particulate matter with a diameter of less than 10 μm (PM₁₀). Ozone was measured in parts per billion (ppb) and PM₁₀ was measured in mass units of micrograms per cubic meter ($\mu\text{g}/\text{m}^3$).

Exposure estimates for air pollution were collected from one monitoring station located in Bloomsbury (central London) via an automated network that undergoes regular quality reviews in accordance with the National Statistics Code of Practice (available from <http://www.statistics.gov.uk>; last accessed on 30 November 2006) and were obtained from the UK National Air Quality Archive (<http://www.airquality.co.uk/archive/index.php>; last accessed on 30 November 2006). The monitoring site was chosen to maximize the period which the data covered and data completeness while maintaining maximal proximity to the location of hospitals in which the births of the SMMIS dataset occurred.

2.4. Statistical analysis

We used regression analysis of daily time-series data to evaluate any associations between the probability of preterm birth (per 1000 fetuses-at-risk) and the various exposure variables. By dividing a series into any trend, seasonal variation, other cyclic change and remaining irregular fluctuations, time-series statistical methods allow for any long-term or slow changes over time (e.g., demographic shifts) to be controlled for while any short-term associations of the health outcome is examined against daily fluctuations of the exposure variables. In practical application, this meant that we were able to account for any known and unknown risk factors that vary slowly across time, but not within individuals, over short periods of time.

First, a core model which was adjusted for seasonality, public holidays and day of week, and between-year variations was created. Then a separate binomial model for each exposure variable was generated.

A parametric approach using Fourier terms was used to recreate a regular seasonal pattern (Stolwijk et al., 1999) and indicator, or 'dummy', variables were used to control for variation in preterm birth between years, day of the week and public holidays. The number of Fourier terms included was determined by the significance of each harmonic and by assessing the deviance, or 'fit', of the model. Optimal seasonal control was achieved by including eight pairs of Fourier terms in the core model, indicating that any regularly occurring patterns or trends beyond approximately 6 weeks were adjusted for. A varied number of Fourier terms (from 4 to 12) was also explored, but did not alter the findings. Therefore, only the results using optimal seasonal control are reported here as this produced the best-fitting overall model.

A plot of the model residuals over time was examined to ensure that no obvious seasonal patterns remained in the preterm birth series. A good fit was demonstrated by a plot of residuals that was randomly distributed with no discernable pattern. The residuals of the core model were also plotted against the predicted values. This plot revealed any model inadequacies as gaps for given ranges of the predicted values or as vertically clumped bands.

We investigated short-term associations using logistic regression and preterm birth probabilities with the number of fetuses-at-risk as the denominator. The number of fetuses-at-risk was calculated by counting any pregnancy that was between 24 weeks (168 days) and less than 37 weeks on the index day. A binomial error structure was specified such that the resulting coefficient represented the log odds ratio per unit increase in exposure. The anti-log of this value was reported in the results, i.e., a coefficient value of greater than one indicated an increased risk of preterm birth and a coefficient value less than one indicated a decreased risk.

We used natural cubic smoothing spline terms to summarize graphically the basic relationship of a preterm birth-exposure association. By dividing each exposure series into equal intervals, constrained to join at the ends, cubic functions allow complex relationships between preterm birth probabilities and any exposure to be modeled without making assumptions about the shape of the potential association. Thus, a flexible, but smooth curve can be produced for visual inspection. The appropriate number of cubic spline terms to include in each model was determined by using Akaike's Information Criterion (AIC) (Chatfield, 2004). The AIC subjects a model to a penalty term that increases as the number of parameters in the model increases, thereby allowing assessment of whether additional complexity is worthwhile and preventing over-fitting. We also used cubic spline terms to control for environmental exposures other than the main variable of interest.

Within the relationships, if a linear association was apparent, the spline terms were replaced with a linear term to quantify the exposure effect. Maximum likelihood estimation was used to identify any threshold in the relationships. Regressions were repeated after varying the threshold by integer values until the best fit, defined by the lowest deviance, was found. The coefficient then represented the increased risk of preterm birth per unit increase in exposure above or below the identified threshold.

2.5. Lag effects of exposures

Any potential effects of weather or air pollution may manifest itself at any time during a pregnancy. Our study examined exposure around the time of birth using lag terms that were calculated by averaging the values of an exposure over a specified period of time. The lag terms were then added to the model in lieu of the exposure value on the day of birth (lag 0). Averaged lag terms for up to 6 days before birth (i.e., day 0–1, day 0–2, day 0–3, day 0–4, day 0–5, day 0–6) were explored in the models for each exposure as previous studies on preterm birth and pollution (Sagiv et al., 2005) or barometric pressure (Cesario, 2002; King et al., 1997), have indicated that any effect of these exposures is likely to occur sometime during the week before birth. When lag terms greater than zero were added to a model, any association between the probability of preterm birth and the exposure of interest were then assumed to represent a cumulative, linear effect over the time frame chosen.

2.6. Autocorrelation

A particular feature of time-series data is that they are correlated through time and not independent. To check for autocorrelation, we inspected the residuals of each model using the partial autocorrelation function (PACF). Significant positive partial autocorrelation coefficients were accounted for by including the appropriate number of autoregressive terms (i.e., lagged residuals) in the model. After the inclusion of the autoregressive terms, the PACF was run again to check that all autocorrelation was accounted for. If autocorrelation still existed, the process was repeated until none remained. No autocorrelation was found in the core model. Autocorrelation was systematically checked as each exposure was added to the core model and controlled for, as necessary.

2.7. Sensitivity analysis

One of the limitations of an ecological study is imprecise exposure assessment as each mother is assumed to be exposed to the same extent. In an effort to account for this, we conducted a sensitivity analysis to investigate any potential effect that distance from the monitoring stations may have had on the results. However, because the data in SMMIS were non-attributable, information on the mother's residence and place of work was not available. Consequently, hospital of birth was used as a proxy. All births that occurred in hospitals located more than 10 miles from the air pollution or weather-monitoring stations were excluded for the sensitivity analysis ($n = 258,341$).

All analysis was carried out using Stata, version 8.2 (StataCorp LP, Texas, USA).

3. Results

Among the 482,765 live, singleton births in SMMIS, 29,716 (6.16%) were preterm. There were a total of 4749 days during the study period with an average of 101.7 (range 52–148) births on each of those days, of which an average of 6.3 (range 0–17) were preterm. There were 13 days (0.27%) on which no preterm births were recorded for the study area. The risk of preterm birth varied significantly between the years ($\chi^2 = 76.06$, $p < 0.001$, 12 d.f.) and on weekends versus weekdays ($\chi^2 = 111.42$, $p < 0.001$). The annual distribution of preterm birth probabilities showed a seasonal variation with a higher incidence during winters than during summers (Fig. 1). There was a 10% increase in risk of being born preterm in winter when compared with summer (relative risk (RR) 1.10, 95% confidence interval (CI) 1.07–1.14).

The yearly pattern of each meteorological and air pollution variable is presented in Figs. 2 and 3. The range of daily means for maximum and minimum temperatures and the mean amount of daily rainfall indicated that the study area is subject to a temperate climate. The largest daily drop in barometric pressure demonstrated a clearer seasonal pattern than the daily mean barometric pressure. Visual inspection of the daily mean levels of ambient PM₁₀ did not show any clear seasonality but PM₁₀ levels were inversely correlated with daily mean levels of ambient ozone (Spearman's rho (ρ) = -0.18 , $p < 0.0001$). Daily mean levels of ambient ozone were positively correlated with daily mean temperature (taken as an average of the daily maximum and minimum values; $\rho = 0.42$, $p < 0.0001$).

3.1. Regression analysis

Upon visual inspection, there was little or no relationship with exposure on the day of birth (lag 0) for any of the meteorological or air pollution variables; spline curves of each exposure measure were similarly flat and therefore only that for daily maximum temperature and the daily mean levels of ambient ozone are shown here (Fig. 4). As each of the relationships was flat and linear across the entire range of values for each of the exposures, the spline

terms were replaced with one linear term consisting of the actual daily values to quantify the association. After adjusting the core model for public holidays, seasonality, and day of the week and between year variations, little or no increase in risk of preterm birth was found with exposure on the day of birth to any of the variables investigated (Table 1). The addition of spline terms to control for daily mean temperature on the day of birth in the models for ozone, daily sunshine, and relative humidity did not alter the findings.

3.2. Delayed effects

When the summed effect of the individual daily lags for up to 6 days before birth were investigated in lieu of exposure on the day of birth, little or no association between preterm birth and any of the exposure variables was found (Fig. 5, left column).

3.3. Sensitivity analysis

Similarly, when the births that occurred in the hospitals located furthest from the monitoring station were removed, there were no differences in associations when compared with the analysis that included all births. Little or no association at any lag was found (Fig. 5, right column).

4. Discussion

There has been increasing interest in the association of reproductive outcomes with environmental factors such as

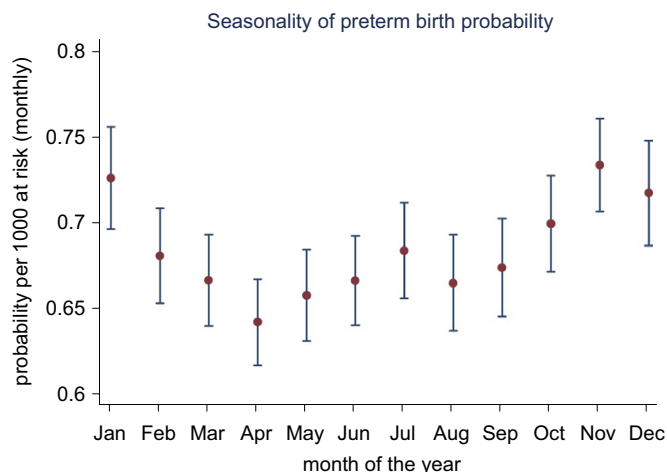


Fig. 1. The seasonal pattern of preterm birth probabilities plotted over one 12-month period. In London-based hospitals, the risk of preterm birth was consistently higher in winters. Vertical bars represent 95% confidence intervals.

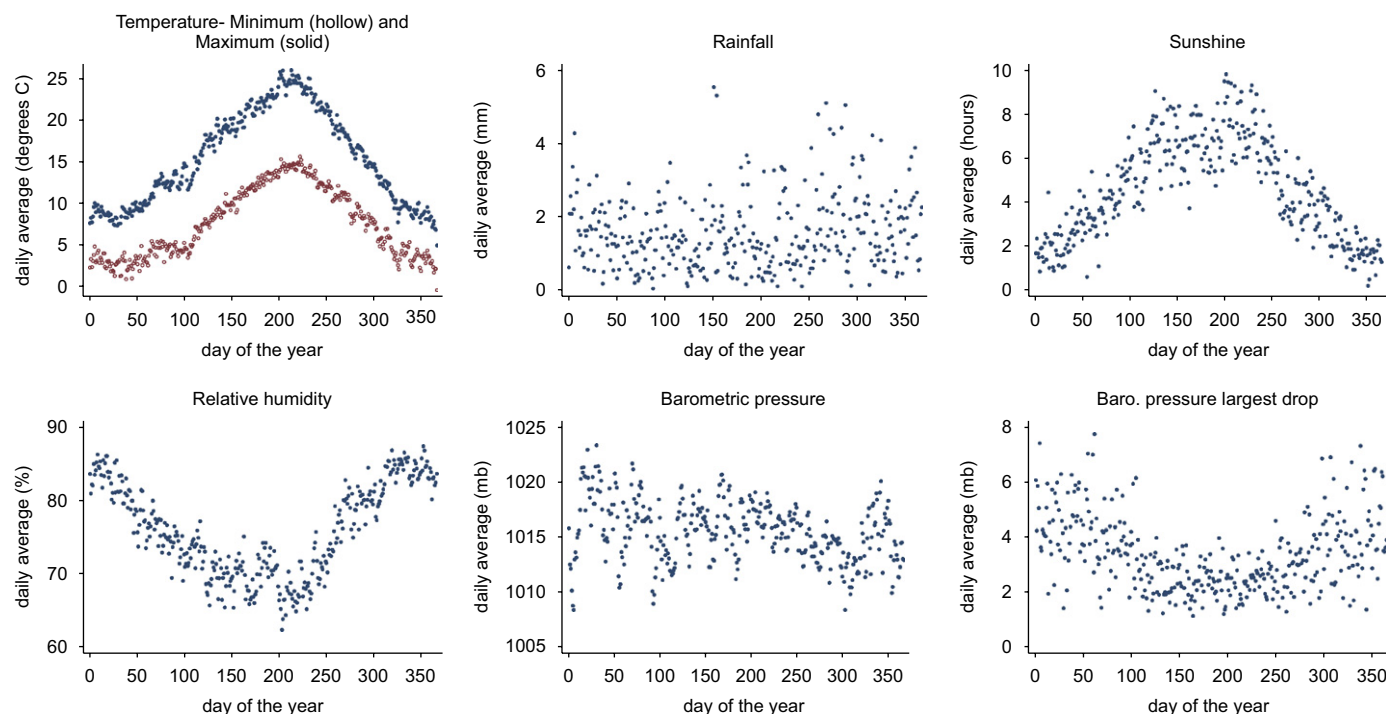


Fig. 2. Annual pattern of climate exposure variables by daily mean.

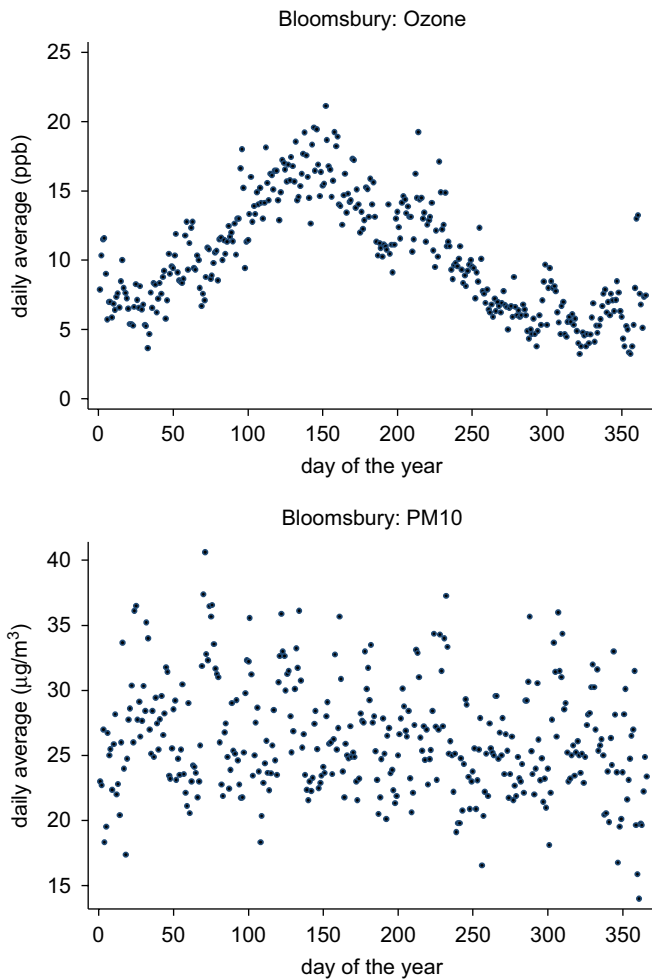


Fig. 3. The annual pattern of ambient ozone and PM₁₀ from the Bloomsbury monitoring station by daily mean in London, UK. Data were only available from 1992 to 2000 for the air pollution exposures.

air pollution, yet studies on air pollution and preterm birth have presented conflicting results (Glinianaia et al., 2004; Maisonet et al., 2004; Sram et al., 2005). Similarly, reports on the association between preterm birth and meteorological factors, such as changes in barometric pressure, have given conflicting findings (Cesario, 2002; King et al., 1997; Noller et al., 1996). Our study analyzed a large dataset and found no increased risk for preterm birth with exposure on the day of birth to daily mean levels of ambient ozone or PM₁₀, daily maximum and minimum temperatures, hours of daily sunshine, the daily amount of rainfall, daily mean barometric pressure or the largest daily drop in barometric pressure. Investigation of cumulative exposure from 0 to 6 days before births also did not show any effect on the risk of preterm birth. In summary, neither the weather nor air pollution affected the time of birth. To our knowledge, this is the first study to investigate any associations between air pollution or climatic factors and preterm birth in the UK using time-series regression techniques.

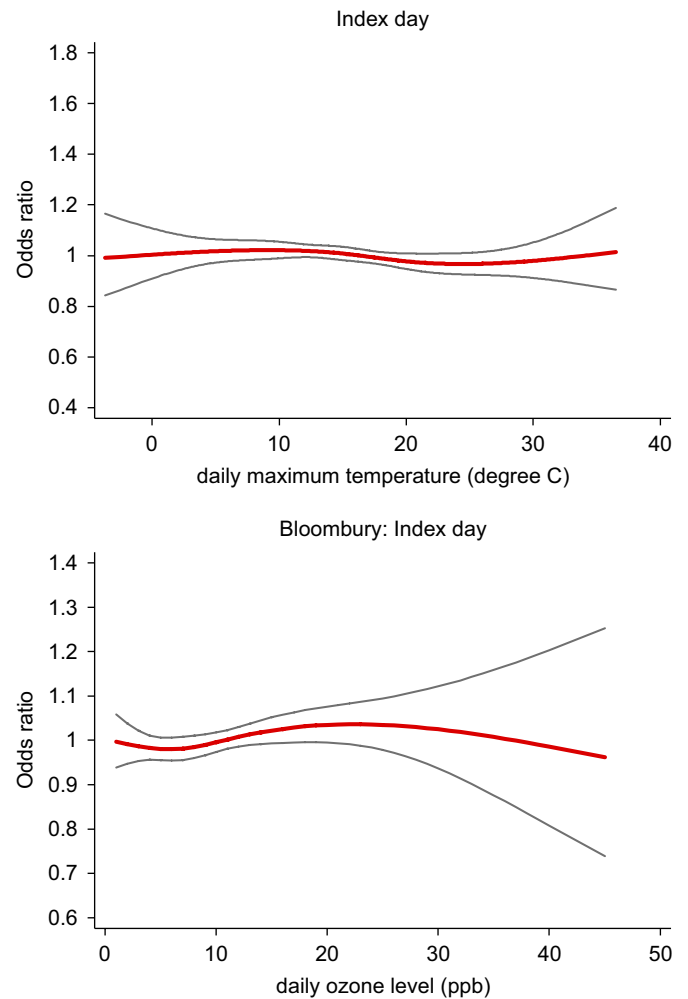


Fig. 4. Little or no increase in risk of preterm birth was found with exposure on the day of birth to any of the variables investigated. The spline curves of each exposure measure were similarly flat, as shown here (with 95% confidence intervals) for daily maximum temperature and the daily mean levels of ambient ozone.

Table 1

Adjusted binomial regression analysis investigating the effect of exposure on the day of birth (index day) on daily preterm birth probability per 1000 fetuses at risk

Exposure	OR ^a	95% CI
Meteorological		
Daily maximum temperature (°C)	1.00	0.99–1.00
Daily minimum temperature (°C)	1.00	1.00–1.00
Daily sunshine (h) ^b	1.00	1.00–1.01
Daily rainfall (mm)	1.00	1.00–1.00
Daily mean barometric pressure (mbar)	1.00	1.00–1.00
Largest daily drop in barometric pressure (mbar)	1.00	0.99–1.00
Daily mean relative humidity (%) ^b	1.00	1.00–1.00
Air pollution		
Daily mean ozone (ppb) ^b	1.00	1.00–1.01
PM ₁₀ (µg/m ³)	1.00	1.00–1.00

^aOdds ratio.

^bIncludes control for daily mean temperature.

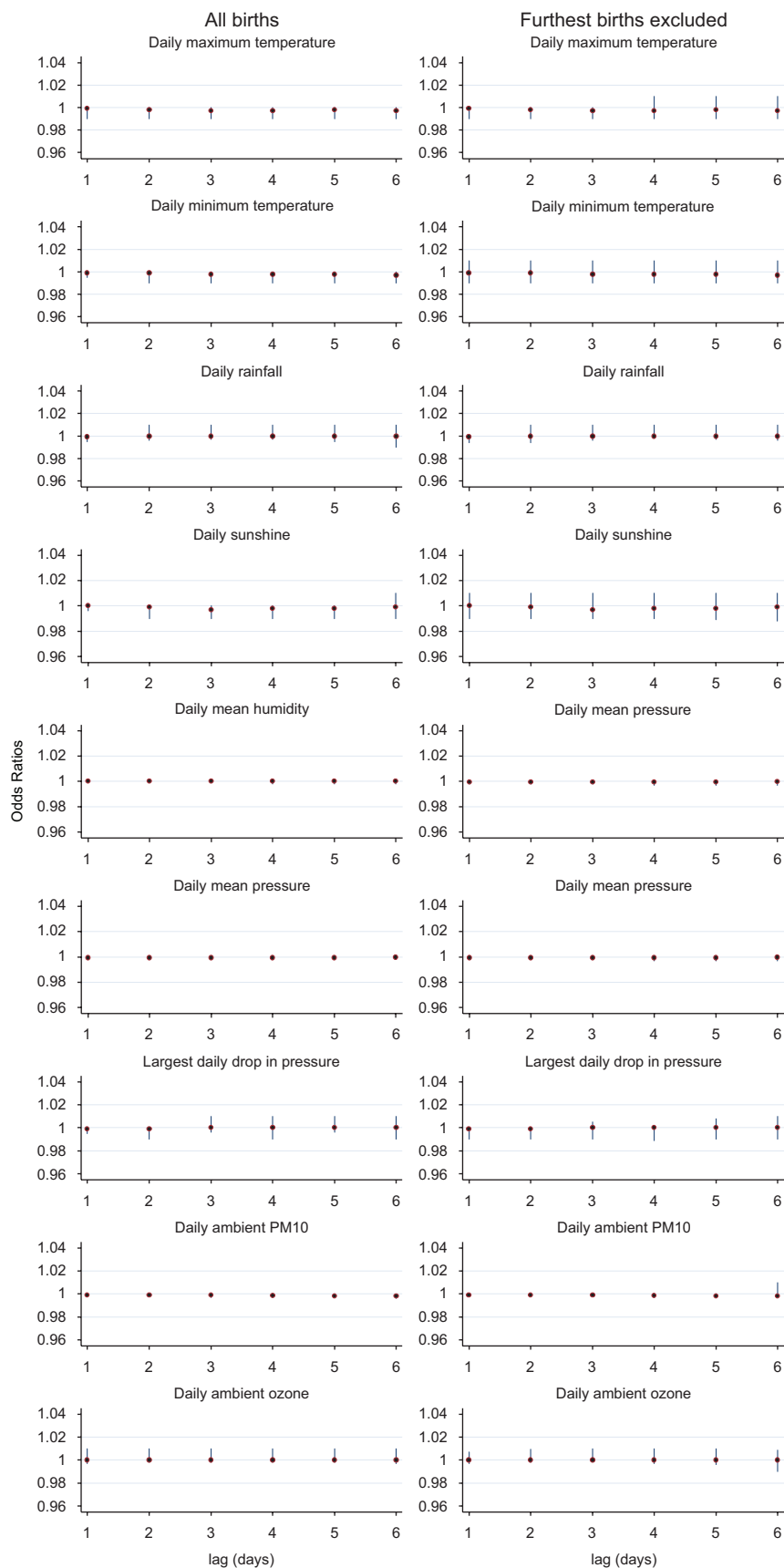


Fig. 5. Left column—odds ratios and 95% confidence intervals for preterm birth and various risk factors, lagged 1–6 days before birth; right column—sensitivity analysis conducted after excluding all births that occurred in hospitals located 10 or more miles away from the monitoring station. Odds ratios and 95% confidence intervals for preterm birth and various risk factors, lagged 1–6 days before birth.

4.1. Other studies and the methodology we used

While an association between air pollution and preterm birth has been reported in other countries such as Croatia, Taiwan, Lithuania, the United States, Canada, the Czech Republic, and China (Alebic Juretic et al., 2001; Bobak, 2000; Liu et al., 2003; Marozienne and Grazuleviciene, 2002; Ritz et al., 2000; Sagiv et al., 2005; Tsai et al., 2003; Woodruff et al., 2003; Xu et al., 1995; Yang et al., 2002a, b, 2003, 2004), this large study and a study conducted in Sweden (Landgren, 1996) found no association between air pollution and preterm birth. Many of the studies, including the Swedish study which reported no association, used a methodology in which the risk of preterm birth in areas of high exposure was compared with areas of low exposure (Landgren, 1996; Tsai et al., 2003; Yang et al., 2002a, b, 2003, 2004). Other studies grouped preterm births across exposure categories according to pollutant concentrations to compare levels of risk (Liu et al., 2003; Marozienne and Grazuleviciene, 2002) or assessed the risk of air pollution as a continuous exposure using linear regression (Bobak, 2000; Ritz et al., 2000). Only two studies used a time-series design (Sagiv et al., 2005; Xu et al., 1995), and only one of these assessed exposure across time, rather than at an individual level (Sagiv et al., 2005). With the exception of this one study, previous studies using spatial distributions of air pollution between individuals have required adjustment for confounding factors, such as maternal smoking and socioeconomic status. For an outcome such as preterm birth, however, such methodologies are unable to account for the many risk factors that remain unknown (Berkowitz and Papiernik, 1993; Kramer, 1987). Time-series regression techniques, which are commonly used for other outcomes of environmental epidemiological research, such as in air pollution and mortality and temperature and mortality studies (Bell et al., 2005; Gouveia et al., 2003; Hajat et al., 2002), have the particular strength of adjusting for the influence of any known and unknown risk factors that vary slowly over time, but not within individuals, over short periods of time.

Furthermore, among the studies that reported an association between air pollution and preterm birth, only three investigated the effects of ozone or PM₁₀ specifically (Sagiv et al., 2005; Liu et al., 2003; Ritz et al., 2000). While Ritz et al. reported a 20% increase in preterm birth per 50 µg increase in ambient PM₁₀ levels averaged over 6 weeks before birth (RR 1.20, 95% CI 1.09–1.33), Sagiv et al. reported a significant increased risk for preterm birth with exposure to average sulfur dioxide exposure in the 6 weeks before birth (RR 1.15, 95% CI 1.00–1.32 per 15 ppb increase) but not for PM₁₀ (RR 1.07, 95% CI 0.98–1.18 per 50 µg increase). Ritz et al. also analyzed the risk of preterm birth associated with ambient ozone and, consistent with our findings, reported no effect. Similarly, while Liu et al. found that preterm birth was associated with exposure to carbon monoxide and also with exposure to sulfur dioxide, they reported no association with exposure to ozone in

either the first month of pregnancy or the last (odds ratio (OR) 0.98, 95% CI 0.89–1.03 and OR 0.93, 95% CI 0.86–1.00, respectively).

We found only one study in the literature reporting a significant association between preterm labor rates and the heat–humidity index (Lajinian et al., 1997). When using preterm birth rather than preterm labor as the outcome, however, the association was no longer significant. In studies of the effect of changes in barometric pressure and labor onset, results have been mixed (Cesario, 2002). Some studies have reported no association (Noller et al., 1996), others report an association with pressure drops (King et al., 1997) and still others report an association when pressure levels are at their highest (Trapasso and Yurchisin, 1988). In general, there was a lack of *a priori* hypotheses of how meteorological factors might influence the risk of preterm birth. It may be that these studies were undersized and therefore not able to properly detect potentially small effects. A particular strength of our study, however, was that the sample size was large and covered a long period of time, so that we can be confident that we have not missed even small effects.

4.2. Exposure mechanisms

Environmental exposures may manifest their effect(s) in at least two ways. First, a relatively immediate effect could result in an early birth soon after exposure or from short-term exposure at some other period during the pregnancy. Second, a cumulative effect after a longer period of exposure throughout (or at various times during) the pregnancy may result in an early birth. This study investigated only the first possibility of a short-term effect occurring soon after exposure. The issue of whether there is a potential for long-term exposures to accumulate and thereby have an effect remains to be addressed.

Although it has been hypothesized that susceptibility to environmental effects in utero may occur only during sensitive and brief periods during development (Bateson et al., 2004), the critical period of exposure to meteorological, air pollution, and infectious agents during pregnancy has not been established. Others have suggested that potential adverse effects may have their largest impact during times of rapid growth of the fetus, such as between 20 and 32 weeks after conception (Colvin et al., 2004).

Complicating the situation even further is the potential for exposure at different times during gestation to have different effects. For example, in a British study of ambient temperature and birthweight (Lawlor et al., 2005), mean temperature during the first trimester was inversely associated with birthweight whereas mean temperature during the third trimester was positively associated with birthweight. The analysis in our study was limited to the period 1 week before birth and investigation of other windows of exposure are not reported here.

4.3. Limitations

Because this was an ecological study, the results are not generalizable at an individual level. Another aspect of an ecological study design is that each mother was assumed to be exposed to the same extent and reliance on ambient measurements from monitoring stations meant that exposure estimates may not be a good reflection of individual exposure for all. Many factors may contribute to differences between area wide and individual level exposures for both air pollution and meteorological variables. For example, spatial distributions of air pollutants may not be homogeneous within an area (e.g., some argue that ozone is less localized when compared with PM₁₀ which has many point sources) or a pregnant woman may spend much of her time indoors (indoor sources of air pollution may also contribute to exposure, further contributing to the difference (Ebelt et al., 2005)), somewhere distant from the monitoring station or she may have recently moved into the study area.

Specifically with regards to meteorological factors, cultural adaptations may result in imprecise exposure assessment. For example, a pregnant woman may spend most of her time indoors in air conditioning when it is hot during the summers and in a heated house (or workplace) when it is cold during winters. Actions that negate the effects of outdoor weather are prevalent in British culture and these may counteract any effect of ambient temperatures on preterm birth.

As with all studies on preterm birth, the measurement of gestational age can be problematic. It has been suggested that errors in estimation of gestational age by LMP may be systematic, in particular, with a tendency to overstate the duration of gestation (Kramer et al., 1988; Savitz et al., 2002). In contrast, early ultrasound scans appear to reduce gestational age estimates when compared with LMP dating and may result in 13–20% higher rates of preterm birth (Yang et al., 2002c; Henriksen et al., 1995). The SMMIS dataset used LMP when available and certain and the ‘best antenatal assessment’ was made using available ultrasound scans when LMP was not available or uncertain. The systematic tendency to overestimate gestational age when using LMP may have resulted in an underestimate of preterm births in this dataset. However, this was likely to be offset by an overestimate of preterm births from ultrasound estimates.

5. Conclusions

Our study provides support to the hypothesis that the daily mean levels of ambient ozone and PM₁₀ that occur in and around London are not associated with an increased risk for preterm birth. Nor are any components of the temperate climate that is experienced in this region associated with an increased risk of preterm birth. One of the strengths of this study was the power provided by the large sample size and the time period covered to detect

small associations, although we were also limited by imprecise exposure assessment. The issue of whether environmental exposures may accumulate over time and thereby have an effect remains to be addressed in future studies.

Many international organizations are now warning about the potential impact that climate change will have on human health (WHO et al., 2003). That the climate is indeed changing rapidly is demonstrated by the fact that 9 of the warmest years on record in Europe have occurred since 1989 and by reconstructions that show unprecedented warming in the last decade (Luterbacher et al., 2004). It is predicted that more intense, more frequent and longer lasting heatwaves will occur during the second half of the 21st century (Meehl and Tebaldi, 2004). A change in temperature and other meteorological factors in Britain could mean a change in their impact on preterm birth. In addition, as air pollution demonstrates seasonal patterns based on climatological conditions, climate change may impact their effect on preterm birth patterns as well. With predictions of climate change and its potential effects on our health, investigation of the effects of weather and air pollution on preterm birth remains relevant.

Acknowledgments

We would like to thank all the maternity units for contributing data to SMMIS, the supportive clinicians and the midwives and clerks who enter the data. We would also like to express our thanks to Noreen Maconochie who provided advice on the analysis.

This work was conducted as part of a self-funded Ph.D. project. This study was reviewed and approved by the London School of Hygiene and Tropical Medicine Ethics Committee as well as the ethics committee at St Mary's Hospital, London, UK.

References

- Alebic Juretic, A., Frkovic, A., Simic, D., 2001. The effect of air pollution on the outcome of pregnancies. *Int. J. Gynaecol. Obstet.* 75 (3), 315–316.
- Anderson, P., Doyle, L.W., 2003. Neurobehavioral outcomes of school-age children born extremely low birth weight or very preterm in the 1990s. *JAMA* 289 (24), 3264–3272.
- Bateson, P., Barker, D., Clutton-Brock, T., Deb, D., D'Udine, B., Foley, R.A., et al., 2004. Developmental plasticity and human health. *Nature* 430 (6998), 419–421.
- Bell, M.L., Dominici, F., Samet, J.M., 2005. A meta-analysis of time-series studies of ozone and mortality with comparison to the national morbidity, mortality, and air pollution study. *Epidemiology* 16 (4), 436–445.
- Berkowitz, G.S., Papiernik, E., 1993. Epidemiology of preterm birth. *Epidemiol. Rev.* 15 (2), 414–443.
- Birth Statistics, 2004. Births and patterns of family building England and Wales (FMI). London, Office for National Statistics.
- Bhutta, A.T., Cleves, M.A., Casey, P.H., Cradock, M.M., Anand, K.J., 2002. Cognitive and behavioral outcomes of school-aged children who were born preterm: a meta-analysis. *JAMA* 288 (6), 728–737.

- Bobak, M., 2000. Outdoor air pollution, low birth weight, and prematurity. *Environ. Health Perspect.* 108 (2), 173–176.
- Byrne, B., Morrison, J., 2004. Preterm birth. *Clin. Evid.* (11), 1903–1922.
- Cesario, S.K., 2002. The ‘Christmas Effect’ and other biometeorologic influences in childbearing and the health of women. *J. Obstet. Gynecol. Neonatal. Nurs.* 31 (5), 526–535.
- Chamberlain, G., Morgan, M. (Eds.), 2002. *ABC of Antenatal Care*, fourth ed. BMJ Publishing Group, London.
- Chatfield, C., 2004. *The Analysis of Time Series*, sixth ed. Chapman & Hall/CRC, Boca Raton, FL.
- Ciconia Health Care Systems Limited, 1998. *An Introduction to SMMIS (St Mary’s Maternity Information System) Manual*. Ciconia Health Care Systems Limited, London.
- Cleary, R., Beard, R.W., Coles, J., Devlin, H.B., Hopkins, A., Roberts, S., et al., 1994. The quality of routinely collected maternity data. *BJOG* 101 (12), 1042–1047.
- Colvin, M., McGuire, W., Fowlie, P.W., 2004. Neurodevelopmental outcomes after preterm birth. *BMJ* 329 (7479), 1390–1393.
- Copper, R.L., Goldenberg, R.L., Creasy, R.K., DuBard, M.B., Davis, R.O., Entman, S.S., et al., 1993. A multicenter study of preterm birth weight and gestational age-specific neonatal mortality. *Am. J. Obstet. Gynecol.* 168 (1 (Part 1)), 78–84.
- Ebelt, S.T., Wilson, W.E., Brauer, M., 2005. Exposure to ambient and nonambient components of particulate matter: a comparison of health effects. *Epidemiology* 16 (3), 396–405.
- Foulder-Hughes, L.A., Cooke, R.W., 2003. Motor, cognitive, and behavioural disorders in children born very preterm. *Dev. Med. Child. Neurol.* 45 (2), 97–103.
- Glinianaia, S.V., Rankin, J., Bell, R., Pless-Mulloli, T., Howel, D., 2004. Particulate air pollution and fetal health: a systematic review of the epidemiologic evidence. *Epidemiology* 15 (1), 36–45.
- Goldenberg, R.L., Rouse, D.J., 1998. Prevention of premature birth. *N. Engl. J. Med.* 339 (5), 313–320.
- Gouveia, N., Hajat, S., Armstrong, B., 2003. Socioeconomic differentials in the temperature–mortality relationship in Sao Paulo, Brazil. *Int. J. Epidemiol.* 32 (3), 390–397.
- Hack, M., Fanaroff, A.A., 1999. Outcomes of children of extremely low birthweight and gestational age in the 1990s. *Early Hum. Dev.* 53 (3), 193–218.
- Hack, M., Flannery, D.J., Schluchter, M., Cartar, L., Borawski, E., Klein, N., 2002. Outcomes in young adulthood for very-low-birth-weight infants. *N. Engl. J. Med.* 346 (3), 149–157.
- Hajat, S., Kovats, R.S., Atkinson, R.W., Haines, A., 2002. Impact of hot temperatures on death in London: a time series approach. *J. Epidemiol. Community Health* 56 (5), 367–372.
- Haram, K., Mortensen, J.H., Wollen, A.L., 2003. Preterm delivery: an overview. *Acta. Obstet. Gynecol. Scand.* 82 (8), 687–704.
- Hediger, M.L., Overpeck, M.D., Ruan, W.J., Troendle, J.F., 2002. Birthweight and gestational age effects on motor and social development. *Paediatr. Perinat. Epidemiol.* 16 (1), 33–46.
- Henriksen, T.B., Wilcox, A.J., Hedegaard, M., Secher, N.J., 1995. Bias in studies of preterm and postterm delivery due to ultrasound assessment of gestational age. *Epidemiology* 6 (5), 533–537.
- King, E.A., Fleschler, R.G., Cohen, S.M., 1997. Association between significant decrease in barometric pressure and onset of labor. *J. Nurse Midwifery* 42 (1), 32–34.
- Kramer, M.S., 1987. Intrauterine growth and gestational duration determinants. *Pediatrics* 80 (4), 502–511.
- Kramer, M.S., McLean, F.H., Boyd, M.E., Usher, R.H., 1988. The validity of gestational age estimation by menstrual dating in term, preterm, and postterm gestations. *JAMA* 260 (22), 3306–3308.
- Lajinian, S., Hudson, S., Applewhite, L., Feldman, J., Minkoff, H.L., 1997. An association between the heat–humidity index and preterm labor and delivery: a preliminary analysis. *Am. J. Public Health* 87 (7), 1205–1207.
- Landgren, O., 1996. Environmental pollution and delivery outcome in southern Sweden: a study with central registries. *Acta. Paediatr.* 85 (11), 1361–1364.
- Lawlor, D.A., Leon, D.A., Davey Smith, G., 2005. The association of ambient outdoor temperature throughout pregnancy and offspring birthweight: findings from the Aberdeen Children of the 1950s cohort. *BJOG* 112 (5), 647–657.
- Lee, S.J., 2005. *The seasonality of preterm birth in a British cohort: a time-series investigation*. Ph.D. Thesis. London School of Hygiene and Tropical Medicine, London.
- Liu, S., Krewski, D., Shi, Y., Chen, Y., Burnett, R.T., 2003. Association between gaseous ambient air pollutants and adverse pregnancy outcomes in Vancouver, Canada. *Environ. Health Perspect.* 111 (14), 1773–1778.
- Luterbacher, J., Dietrich, D., Xoplaki, E., Grosjean, M., Wanner, H., 2004. European seasonal and annual temperature variability, trends, and extremes since 1500. *Science* 303 (5663), 1499–1503.
- Maisonet, M., Correa, A., Misra, D., Jaakkola, J.J., 2004. A review of the literature on the effects of ambient air pollution on fetal growth. *Environ. Res.* 95 (1), 106–115.
- Marlow, N., Wolke, D., Bracewell, M.A., Samara, M., 2005. Neurologic and developmental disability at six years of age after extremely preterm birth. *N. Engl. J. Med.* 352 (1), 9–19.
- Marozienne, L., Grazuleviciene, R., 2002. Maternal exposure to low-level air pollution and pregnancy outcomes: a population-based study. *Environ. Health* 1 (1), 6.
- Meehl, G.A., Tebaldi, C., 2004. More intense, more frequent, and longer lasting heat waves in the 21st century. *Science* 305 (5686), 994–997.
- Noller, K.L., Resseguie, L.J., Voss, V., 1996. The effect of changes in atmospheric pressure on the occurrence of the spontaneous onset of labor in term pregnancies. *Am. J. Obstet. Gynecol.* 174 (4), 1192–1197.
- Office for National Statistics, 2004. *Birth Statistics: Births and patterns of Family Building England and Wales (FM1)*. Office for National Statistics, London.
- Petrou, S., 2003. Economic consequences of preterm birth and low birthweight. *BJOG* 110 (Suppl. 20), 17–23.
- Petrou, S., 2005. The economic consequences of preterm birth during the first 10 years of life. *BJOG* 112 (Suppl. 1), 10–15.
- Porter, K.R., Thomas, S.D., Whitman, S., 1999. The relation of gestation length to short-term heat stress. *Am. J. Public Health* 89 (7), 1090–1092.
- Ritz, B., Yu, F., Chapa, G., Fruin, S., 2000. Effect of air pollution on preterm birth among children born in Southern California between 1989 and 1993. *Epidemiology* 11 (5), 502–511.
- Romero, R., Espinoza, J., Kusanovic, J.P., Gotsch, F., Hassan, S., Erez, O., Chaiworapongsa, T., Mazar, M., 2006. The preterm parturition syndrome. *BJOG* 113 (s3), 17–42.
- Sagiv, S.K., Mendola, P., Loomis, D., Herring, A.H., Neas, L.M., Savitz, D.A., et al., 2005. A time-series analysis of air pollution and preterm birth in Pennsylvania, 1997–2001. *Environ. Health Perspect.* 113 (5), 602–606.
- Savitz, D.A., Terry Jr., J.W., Dole, N., Thorp Jr., J.M., Siega Riz, A.M., Herring, A.H., 2002. Comparison of pregnancy dating by last menstrual period, ultrasound scanning, and their combination. *Am. J. Obstet. Gynecol.* 187 (6), 1660–1666.
- Sram, R.J., Binkova, B., Dejmek, J., Bobak, M., 2005. Ambient air pollution and pregnancy outcomes: a review of the literature. *Environ. Health Perspect.* 113 (4), 375–382.
- Stolwijk, A.M., Straatman, H., Zielhuis, G.A., 1999. Studying seasonality by using sine and cosine functions in regression analysis. *J. Epidemiol. Community Health* 53 (4), 235–238.
- Trapasso, L.M., Yurchisin, M.J., 1988. Weather and childbirth: is there a link? *Female Patient* 13, 33–37.
- Tsai, S.S., Yu, H.S., Liu, C.C., Yang, C.Y., 2003. Increased incidence of preterm delivery in mothers residing in an industrialized area in Taiwan. *J. Toxicol. Environ. Health A* 66 (11), 987–994.
- Tucker, J., McGuire, W., 2004. *ABC of preterm birth: epidemiology of preterm birth*. *BMJ* 329 (7467), 675–678.
- Villar, J., Abalos, E., Carroli, G., Giordano, D., Wojdyla, D., Piaggio, G., et al., 2004. Heterogeneity of perinatal outcomes in the preterm delivery syndrome. *Obstet. Gynecol.* 104 (1), 78–87.

- WHO, 1972. The prevention of perinatal morbidity and mortality: report on a seminar. Public Health Papers No. 42. World Health Organization, Geneva.
- WHO WMO, UNEP, 2003. Climate Change and Human Health—Risks and Responses. Summary. World Health Organisation, Geneva.
- Woodruff, T.J., Parker, J.D., Kyle, A.D., Schoendorf, K.C., 2003. Disparities in exposure to air pollution during pregnancy. *Environ. Health. Perspect.* 111 (7), 942–946.
- Xu, X., Ding, H., Wang, X., 1995. Acute effects of total suspended particles and sulfur dioxides on preterm delivery: a community-based cohort study. *Arch. Environ. Health* 50 (6), 407–415.
- Yang, C.Y., Cheng, B.H., Hsu, T.Y., Chuang, H.Y., Wu, T.N., Chen, P.C., 2002a. Association between petrochemical air pollution and adverse pregnancy outcomes in Taiwan. *Arch. Environ. Health* 57 (5), 461–465.
- Yang, C.Y., Chiu, H.F., Tsai, S.S., Chang, C.C., Chuang, H.Y., 2002b. Increased risk of preterm delivery in areas with cancer mortality problems from petrochemical complexes. *Environ. Res.* 89 (3), 195–200.
- Yang, H., Kramer, M.S., Platt, R.W., Blondel, B., Breart, G., Morin, I., et al., 2002c. How does early ultrasound scan estimation of gestational age lead to higher rates of preterm birth? *Am. J. Obstet. Gynecol.* 186 (3), 433–437.
- Yang, C.Y., Chang, C.C., Chuang, H.Y., Ho, C.K., Wu, T.N., Tsai, S.S., 2003. Evidence for increased risks of preterm delivery in a population residing near a freeway in Taiwan. *Arch. Environ. Health* 58 (10), 649–654.
- Yang, C.Y., Chang, C.C., Chuang, H.Y., Ho, C.K., Wu, T.N., Chang, P.Y., 2004. Increased risk of preterm delivery among people living near the three oil refineries in Taiwan. *Environ. Int.* 30 (3), 337–342.